

SunoBolo — Voice-Based Inventory Management Using Conversational Speech

Milestone 1.1 · Requirements Gathering & Analysis

Deliverable: Requirements Document

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Prepared by: _____

Project Context

SunoBolo is a voice-first inventory system for small Indian retail shops.

Every voice inventory product on the market works as a **command interface**: you press a button and issue a structured instruction — *"add twenty kilo rice."* That is still data entry, performed with your mouth instead of your thumbs. SunoBolo inverts this. The phone sits on the shop counter and **listens to the shop**, extracting stock movements from the ordinary conversation between owner and customer — sentences that were never addressed to the app at all. A kirana shop already says every transaction out loud, usually twice: once when the customer asks and once when the owner confirms. That speech is a complete, free transaction log that currently evaporates into the air. **SunoBolo captures it.**

Because passive capture is less accurate than dictated commands, nothing it extracts is written to stock automatically. Every extraction becomes a **proposal** the owner approves in bulk, and a conventional **Command Mode** mic button remains available as the reliable path. The novel capability is genuinely novel; the failure mode is an ordinary voice inventory app.

Milestone 1 deliverables

- **1.1** — Requirements Gathering & Analysis (*this document*)
- **1.2** — Functional Requirements
- **1.3** — Technical Design & Architecture
- **1.4** — UI/UX Design & Prototype

Requirements Gathering & Analysis

1. Problem Context

India has roughly 13 million kirana and small trade shops. The overwhelming majority track stock in one of three ways: a paper notebook (*bahi khata*), the owner's memory, or ad-hoc WhatsApp messages to a supplier. All three fail identically — **the record is not queryable.** The owner cannot answer "how much rice is left?" without walking to the godown, and cannot answer "what do I need to order?" at all until something has already run out.

Existing digital inventory software does not solve this, because it moves the work to a keyboard. It assumes the operator can type, can type in English, can read terms like *SKU*, *variant*, and *reorder point*, and has patience for a multi-field form per entry. An owner logging 40 stock movements a day will not do this.

Voice-command inventory apps are a partial fix, and they are already widely available.

They remove the keyboard but keep the *interruption*: the owner must stop serving, open the app, tap a button, and speak a well-formed sentence. In a shop with a queue at the counter, that interruption is the reason the feature goes unused after week one.

The observation that drives this project: the transaction has already been spoken aloud, in natural language, by two people, before any app was opened. The data-entry step is redundant. The system should listen instead of asking.

2. Target Users

Persona	Profile	Primary need
Ramesh, 48 — kirana owner, Tier-2 city	8th standard, fluent Hindi, minimal English, Android phone, uses WhatsApp voice notes daily	Stock to be recorded <i>without stopping work</i>
Lakshmi, 35 — provisions store, Telangana	Speaks Telugu with English product names mixed in, reads numerals but not English words	Know what is running out before customers ask
Arjun, 22 — owner's son, does the	Comfortable with apps	A reliable reorder list, not a guess

ordering		
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Common denominator: all three are fluent *speakers*, reluctant *typists*, and — critically

— already narrate every sale out loud as part of doing business.

3. Evidence Basis

Requirements derive from the GigPoint problem statement and three observable behaviours:

1. **The transaction is spoken before it is recorded.** A counter sale is a verbal exchange:

the customer names the item and quantity, the owner confirms. This happens 100% of the time, with no prompting and no app.

2. **Speech is code-switched, not monolingual.** A real utterance is "*do carton Maggi dena*"

— a Hindi numeral, an English trade unit, a brand name, a Hindi verb. Any system demanding one language per sentence will be abandoned.

3. **Trade units are the mental model.** Stock is thought of in *bori*, *peti*, *dozen*, *tin*,

quintal. Forcing entry in kilograms makes the owner do arithmetic — exactly the friction being removed.

4. Current-State Pain → Requirement Mapping

Pain observed	Consequence	Requirement generated
Entry requires typing	Stock logged late or never	R1 — voice as primary input
Even voice entry interrupts serving	Feature abandoned after initial novelty	R2 — passive capture; zero interaction in the happy path
Software is English-only	Owner needs a helper to operate it	R3 — regional + mixed-language input
Units forced to kg/pieces	Mental arithmetic on every entry	R4 — accept trade units, convert internally
No visibility of totals	Shortages found at point of sale	R5 — at-a-glance dashboard
No reorder signal	Over-ordering and stockouts together	R6 — threshold alerts + reorder list
Speech recognition is imperfect	A wrong entry silently corrupts stock	R7 — nothing auto-commits; propose, then bulk-approve
Notebook can be lost	Total history loss	R8 — durable server-side persistence + history

5. Scope

In scope (v1, delivered today)

- **Counter Mode** — continuous passive listening that proposes stock movements from

overheard conversation

- **Command Mode** — a mic button for deliberate, single, well-formed entries
- Hindi, Telugu and Indian-English input, freely mixed within one sentence
- Trade-unit entry with per-item conversion to a base unit
- Pending-proposal tray with bulk accept/reject
- Stock dashboard, low-stock alerts, reorder suggestions in purchase units
- Spoken and written confirmation in the owner's language
- Full transaction history retaining the original transcript per entry
- Server-side persistence

Out of scope (explicitly deferred)

Billing and GST invoicing · customer credit (*udhaar*) ledger · multi-shop and staff roles ·

barcode or image entry · offline-first sync · demand forecasting beyond a threshold rule ·

supplier ordering integration · speaker diarisation (telling owner and customer apart)

6. Constraints

Constraint	Consequence for design
Zero budget	No paid speech APIs; browser-native recognition only
No model training or fine-tuning	Recognition off-the-shelf; understanding rule-based and fuzzy-matched
Single developer, one working day	Narrow and vertically complete, not broad
Low-digital-literacy users	No software jargon in any visible string; 48 px minimum touch targets
Low-end Android, patchy network	Mobile-first, small payloads, graceful degradation to typed input
Counter device is on a charger	Continuous listening is acceptable; a pocket app it is not

7. Assumptions

4. The shop has a smartphone with Chrome (Android) or Safari (iOS) and intermittent internet.
5. A shop stocks 20–100 distinct items — small enough for exhaustive matching against a

per-shop catalogue rather than open vocabulary.

6. Counter conversations are audible to a phone placed on the counter.
7. Pack sizes (*1 bori rice = 50 kg*) are configured once at setup and change rarely.
8. The owner will review a batch of proposals once or twice a day, rather than confirm each

entry in the moment.

8. Risks

Risk	Likelihood	Impact	Mitigation
Passive extraction accuracy is low	High	High	Nothing auto-commits. Proposals go to a review tray; a wrong proposal

			costs one tap to dismiss, not a corrupted ledger
Conversation contains no usable transaction	High	Low	Low-confidence output is discarded silently rather than shown
Continuous recognition stops on silence or tab blur	High	Medium	Restart-on-end loop; a visible "listening" indicator so the owner can see it died
Browser lacks Web Speech API	Medium	High	Editable transcript field feeds the identical parse endpoint
Privacy objection to listening at the counter	Medium	High	Recognition runs in the browser; no audio is recorded, stored or uploaded — only extracted (<i>item</i> , <i>qty</i> , <i>direction</i>) values
Battery drain / constant network	High	Medium	Positioned as a counter-side device on a charger
Ambiguous unit ("1 bag")	Medium	Medium	Pack size stored per item; unknown unit triggers a clarification chip

9. Success Criteria

On a live phone against the deployed URL:

9. A performed counter conversation produces at least one correct stock proposal without any button being pressed.
10. A Telugu–English mixed sentence is parsed correctly for item, quantity and unit.
11. A trade unit ("*do bori*") is stored correctly converted to the base unit.
12. Every proposal can be accepted or corrected without opening the keyboard.
13. "What is running low?" returns a correct, spoken answer.
14. Stock totals survive a page reload and a server restart.

10. Non-Goals

This is **not** a general-purpose voice assistant. It deliberately understands a closed

domain — the items in one shop's catalogue plus a fixed set of trade units and actions. That

narrowness is precisely why it can be accurate without any trained model, and it is a design

decision rather than a limitation of the timeline.